INCH-POUND

MIL-DTL-32235/2
15 September 2010

DETAIL SPECIFICATION SHEET

HEATER MODULE,
TYPE II: HEATER, ASSEMBLED

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32235.

CLASSIFICATION

Heater modules are of the following styles:

Style A	-	Three sub-units plus a Boil-In-Bag (BIB) module
Style B	-	Four sub-units

REQUIREMENTS

I. Heater module.

The Type II heater module shall consist of a heater module box containing three or four heating trays, activation fluid units, heaters, polymeric food trays or Institutional Size Pouches (ISP) and the activation mechanisms/pull tabs. Sub-assemblies are allowed. When applicable, there shall also be a Boil-In-Bag (BIB) module. The components and sub-units shall be assembled and the activation mechanisms/pull tabs shall be connected to the three or four activation fluid units and to the three or four heaters. The sub-units and the BIB module (when applicable) shall be placed into the heater module box.

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A. Heater.

The heater shall be constructed of materials that, when activated by a fluid, shall initiate and propagate an exothermic reaction suitable for use with food. This reaction shall generate adequate heat to heat the food to a safe food serving temperature. No toxic gas, liquid or solid by-products are desirable. If toxic by-products are produced, they shall be of the least severity and smallest amount possible while allowing for adequate heating and ensuring operator and consumer safety. The heater material shall be evenly distributed and completely sealed within the scrim matrix of the heater to minimize the release of materials, and facilitate direct in-place activation of the heater materials. The heating rate shall be optimized to minimize the time required to heat the food, yet not cause excessive foaming or uncontrolled release of reaction by-products. The heater and barrier material shall not melt, deform or degrade during heating.

The heater is activated by the addition of a fluid that shall fully activate the heater material. The non-woven porous polymeric scrim shall be sealed and sized to accommodate proper fit and function of the heater module. Each heater (heater elements in a matrix) shall be correctly and legibly labeled in accordance with MIL-DTL-32235.

The heater shall be placed in the bottom of the heating tray, covered and hermetically sealed in place with a layer of barrier material to the bottom of the tray with a peelable seal. The barrier material shall be scored (laser or mechanical) and shall be attached to a pull strip to enable activation.

B. Activation fluid unit.

The unit consisting of the pouch containing the activation fluid shall be made of material equivalent to Class 1 of MIL-PRF-131. Alternate activation fluid pouch materials and designs shall be permitted with approval from Natick Soldier Research, Development and Engineering Center. The pouch shall be manufactured in accordance with the dimensions and design shown in Figure 1. Tolerances for the pouch dimensions shall be $\pm$ 1/8 inches. Sufficient length for the center strip and careful assembly is critical to ensuring that the pouch is not inadvertently torn open during assembly and subsequent transport and storage. The solid lines shown at 1 inch off center at the base of the strip are cut lines. The 1 inch center strip section of the pouch shall be constructed with additional material for reinforcement. The center section of the pouch shall be scored (laser or mechanical) to provide easy tear properties without degrading the strength and barrier properties of the pouch. The pouch shall be filled with 1.5 percent saline (water and sodium chloride) solution, or as specified by the heater manufacturer with approval from Natick Soldier Research, Development and Engineering Center. The volume of fluid in the pouch, when combined with the heater, shall be adequate to initiate and propagate the exothermic reaction. Each activation fluid unit shall be correctly and legibly labeled in accordance with MIL-DTL-32235.

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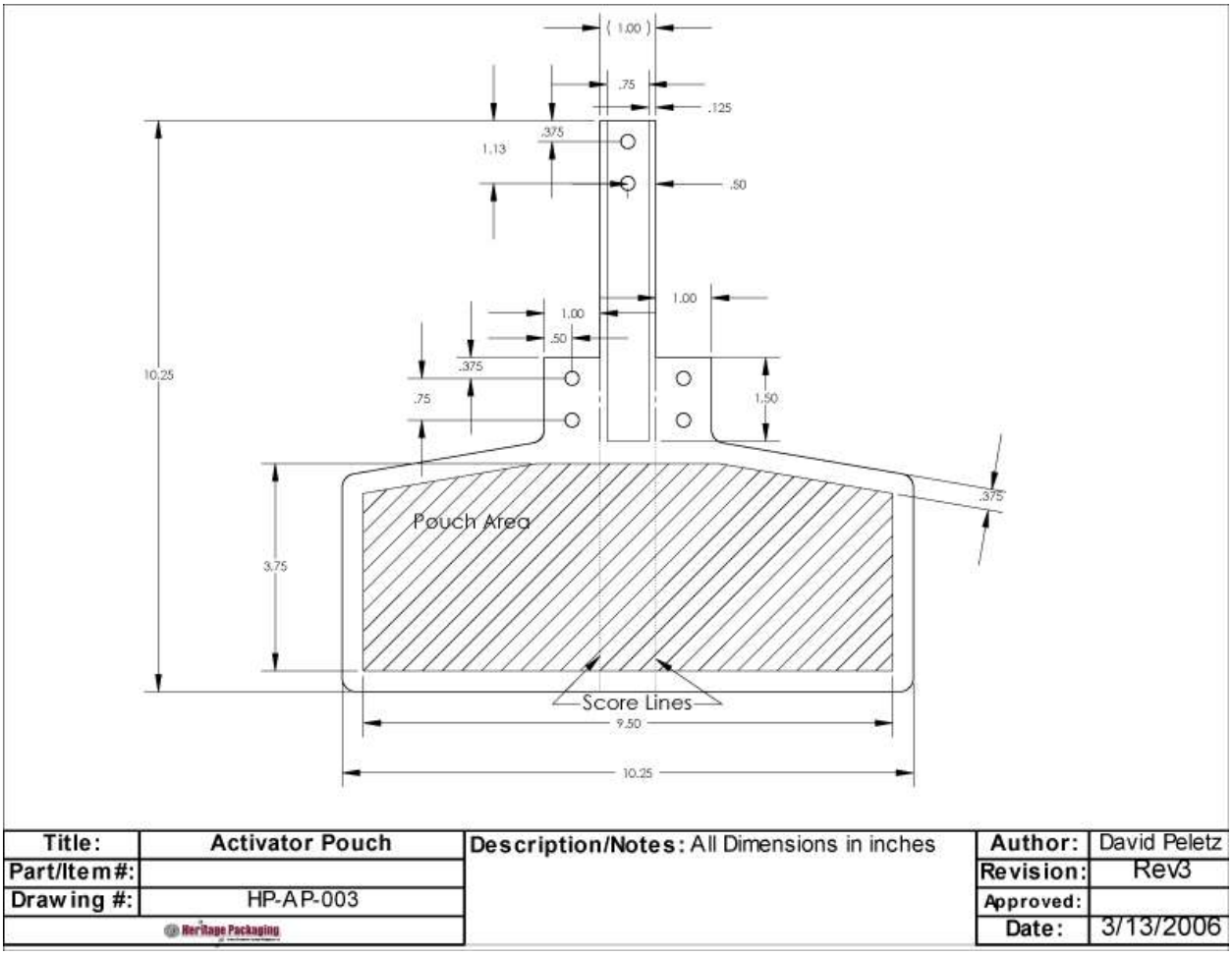


FIGURE 1. Activation fluid unit

C. Heating tray.

The heating tray shall be molded from 0.090 inch high density polyethylene. Injection molded and thermoformed designs have performed adequately; other acceptable processes and materials are allowed. The heating tray shall consist of two compartments. The smaller compartment serves as a retaining reservoir for the activation fluid unit and includes two buttons to which the two short tabs of the activation fluid unit pouch are secured.

A fold-over flap shall be included on the heating tray to retain and protect the activation fluid unit. A slot shall be cut into the cover to allow the center strip of the activation fluid unit pouch to slide through, which later is attached to an activation mechanism/pull tab.

The larger compartment shall be configured to hold the heater, one polymeric food tray or ISP, and accommodate the activation fluid. The compartment shall be configured with a raised and rounded edge that supports the polymeric food tray securely above the heater.

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Dimensions of the heating tray shall be as specified in Figure 2. The tolerance for the angle measurements shall be ± 1 degree. The tolerance for the linear measurements shall be ± 1/8 inch.

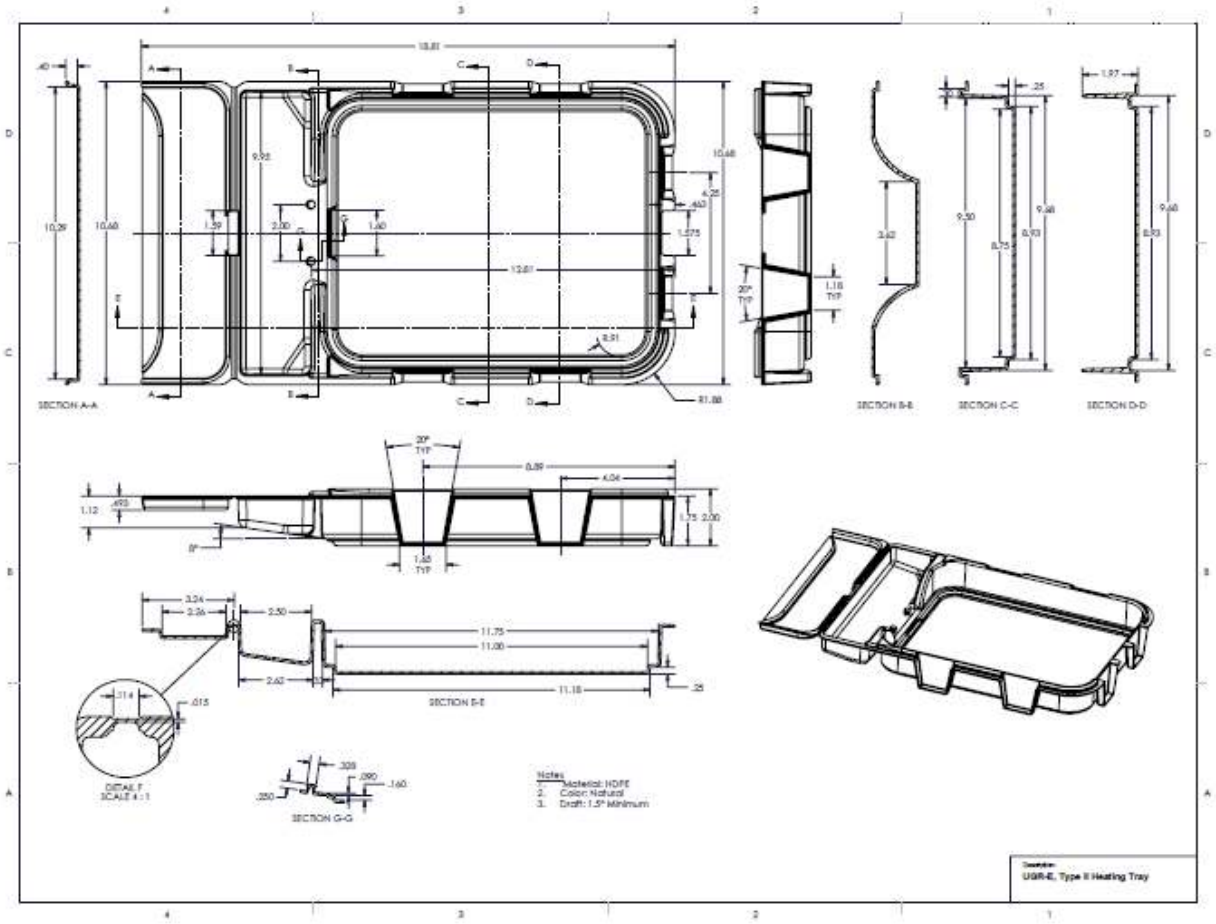


FIGURE 2. Heating tray

D. Pull tabs.

The pull tabs shall be constructed of a blend of low to medium density polyethylene or equivalent material and shall provide high strength characteristics under a wide range of environmental conditions. The material shall withstand temperatures ranging from -20°F to 160°F without fracture or failure. Dimensions of the pull tabs shall be as specified in Figures 3 or 4.

The pull tabs shall be configured with loading stations to support the three or four tray two-step activation method intended for the heater module. Each station shall be configured to retain the assembled activator strip and withstand a minimum pull force of 75 pounds. For a tab configured like the R16-4WCSHCERDG tab (Figure 3), the center strip of the activation fluid unit shall be inserted through the back opening of the tooth and each of the two holes on the extending end of the activator tab are pressed onto this tooth until the edge of the strip is secured

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under both retaining teeth. For a tab configured like the RND-PT tab (Figure 4), the end of the center strip of the activating fluid unit shall be folded between the retaining holes and the tab inserted through the retaining station(s). The second pull tab shall be assembled to the heater pull strips in a similar configuration.

Upon completion of the heater module, the three or four center strips of the activation fluid units securely connect to a pull tab, and the three or four pull strips of the packaged heater securely connect to another pull tab. At the time of use, the operator of the heater module pulls the pull tabs to expose the heaters and to tear the activation fluid unit pouches which open and release the saline solution and activate the heaters.

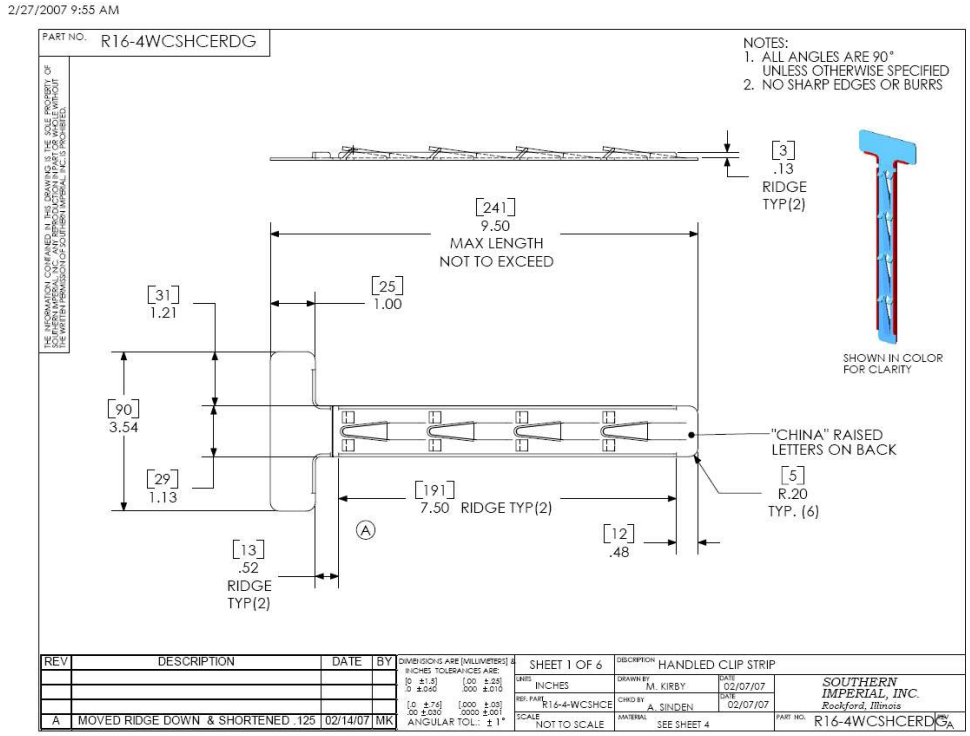


FIGURE 3. Pull tab, Rear loading design

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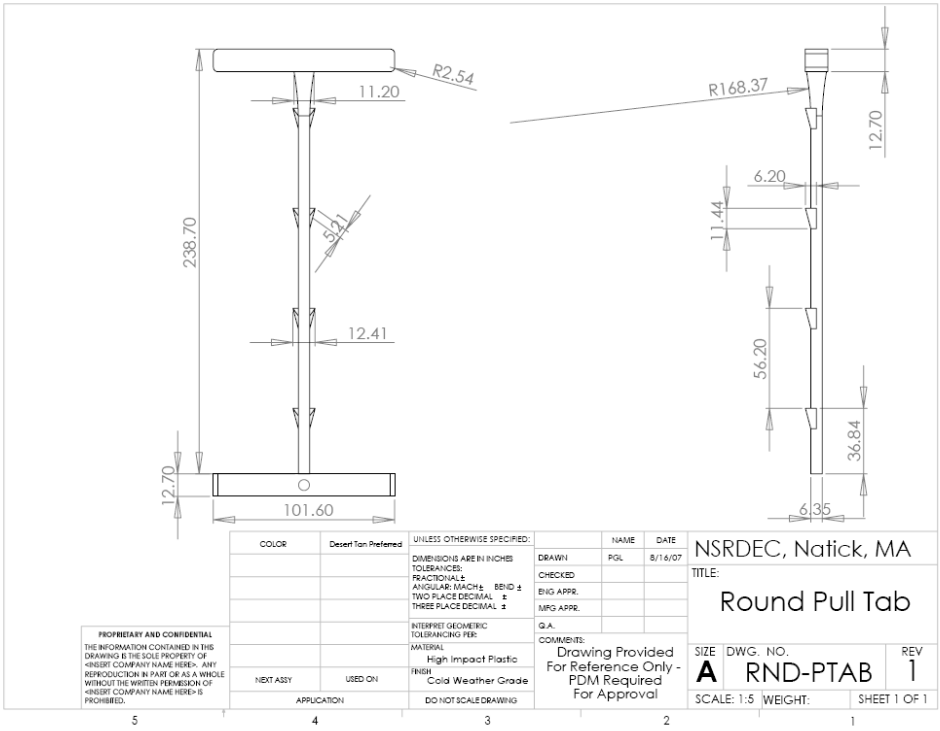


FIGURE 4. Pull tab, Center loading design

E. Assembly of heater module.

The components or sub-units shall be assembled. The Type II heater module shall consist of a box containing three sub-units and a BIB module or four sub-units, with activation mechanisms/pull tabs. The sub-unit shall consist of the heating tray with sealed-in heater, the activation fluid unit and the polymeric food tray or (ISP). Sub-assemblies of components may be used. The three or four sub-units shall be stacked and one activation mechanism/pull tab shall be connected to the three or four activation fluid unit center strips and the other activation mechanism/pull tab shall be connected to the three or four heater pull strips. The sub-units and the BIB module (when applicable) shall be placed into the heater module box. The entrée polymeric food tray shall be on the bottom and the dessert polymeric food tray shall be on the top. When there is a BIB module, it shall be placed on top of the three sub-units.

Corrugated fiberboard pads or cushioning shall be packed as necessary to fill excess headspace. The heater module box shall be closed and instruction sheets applied. Design and dimensions of the heater module box shall be as specified in Figure 5. Each heater module shall be correctly and legibly labeled in accordance with MIL-DTL-32235.

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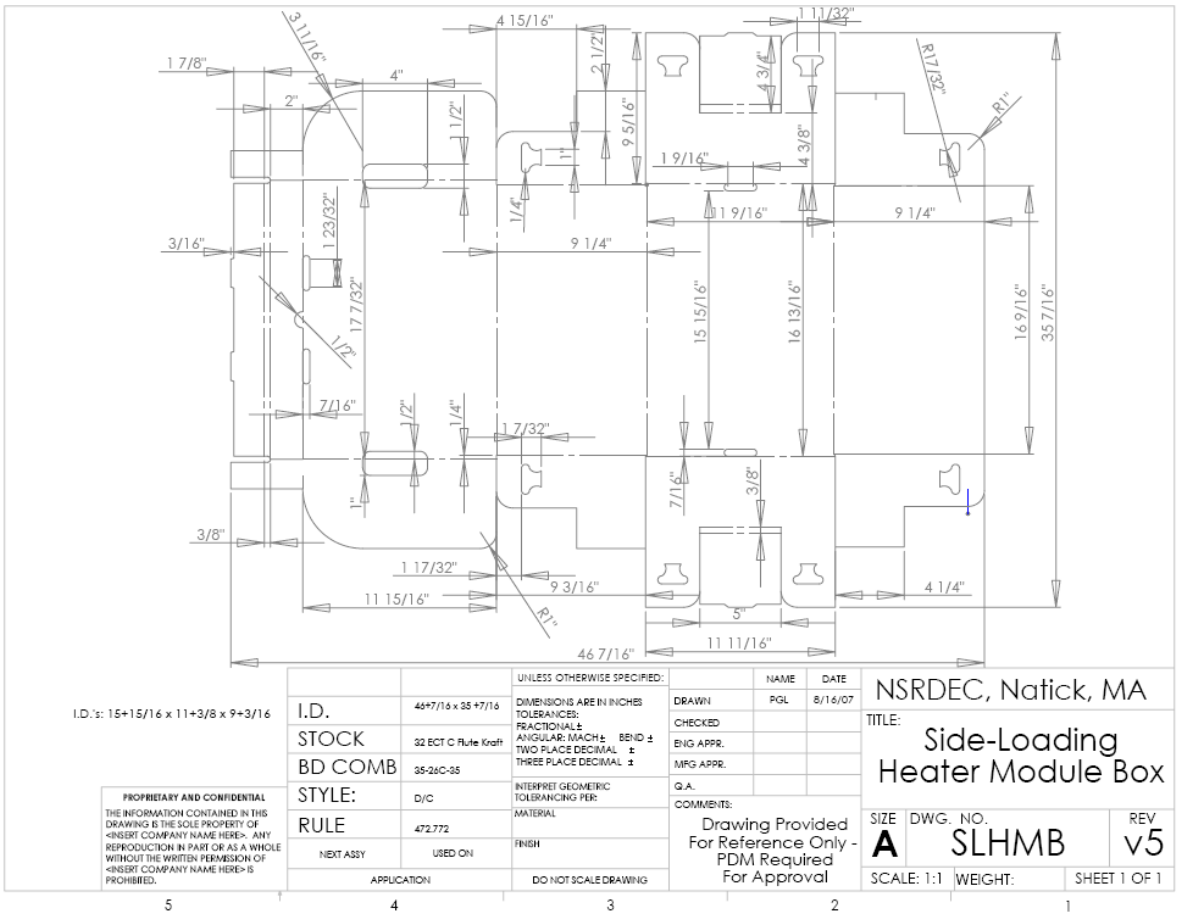


FIGURE 5. Heater module box

EXAMINATION AND TESTS:

A. Type II heater module examination. In addition to the heater module examination specified in MIL-DTL-32235, the finished product shall be examined for conformance to the requirements specified in this specification sheet. The Type II heater module shall be examined for the defects listed in Table I.

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TABLE I. Type II heater module defects 1/ 2/ 3/ 4/ 5/ 6/

Category		Defect
Major	Minor	
101		Heater module not type or not style specified.
102		Heater does not contain materials that will initiate and propagate an exothermic reaction.
103		Heater causes excessive foaming or uncontrolled release of reaction by-products.
104		Heater not hermetically sealed in tray with barrier material.
105		Center strip of activation fluid units not reinforced.
106		Score lines on activation fluid units missing or damaged.
	201	Score line on heater barrier material not provided.
	202	Heater barrier material not connected to pull strip.
	203	Heater pull strip not attached to activation mechanism/pull tab.
107		Activation mechanism/pull tab missing or damaged.
	204	Activation fluid unit pull strip not attached to activation mechanism/pull tab.
	205	Activation mechanism/pull tab not assembled properly.
108		Fold-over flap on heating tray to retain the activation fluid unit missing or damaged.
109		Heating tray design or dimensions not correct.
	206	Polymeric trays or ISPs or BIB, if applicable, of food not placed in module in correct order.
110		Excess headspace in module allowing movement of components.

1/ Heater material construction shall be verified by Certificate of Conformance (CoC).

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2/ Material in accordance with MIL-PRF-131 for the activation fluid unit shall be verified by CoC.

3/ The activation fluid solution shall be identified and verified by CoC.

4/ The use of 0.090 inch high density polyethylene for the heating tray shall be verified by CoC.

5/ The pull tab material shall be verified by CoC.

6/ The pull tab pull strength shall be verified by CoC.

B. Test methods.

1. Single heater capacity test.

The objective of the heating capacity test is to verify that a single heater increases the temperature of the water in the polymeric tray by 85°F (from 40°F to 125°F) in 30 minutes or less. In this test, one sub-unit (polymeric tray of water in the heating tray with the heater and the activation fluid unit) is tested. The following procedures are recommended:

- Pre-condition 96 oz. water-filled test tray to 35°F to 40°F.
- Align matching bi-metallic (copper-constantan) pegs of C-10 Locking Connector to bi-metallic holes in C-9 Locking Receptacle. Thread C-10 Locking Receptacle and C-9 together until seated.
- Connect Thermocouple wire installed on Locking Receptacle to data acquisition or computer terminal calibrated to the copper-constantan thermocouple.
- Assemble heater and food tray within the heating tray. Add activation fluid or use activator pouch to activate heater.
- The test shall be conducted at an ambient temperature of 72°F ± 2°F in an explosion-proof exhaust fume hood or sufficiently ventilated environment, away from open flame or potential ignition sources.
- Place sub-unit into a representative heater module box with 10-3/4 by 16-1/4 inch corrugated insert placed over the tray (weather grade corrugated or plastic materials may be used for repeated testing). Activate the sub-unit.
- Record temperature for at least 30 minutes at 1 minute intervals or more frequently.

II. Notes.

A. Part identifiers and sources of supply.

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1. Heater. The heater is available from:

Truetech Inc.
680 Elton Ave.
Riverhead, NY 11901-2585
(631) 727-8600

2. Heater barrier pouch. The barrier pouch material is available from:

Winter-Wolff International
131 Jericho Turnpike
Jericho, NY 11753
(516) 997-3300

3. Activation fluid unit. The activator pouch is identified as Part # HP-AP-003. The material CADPACK N for the construction of the activation fluid unit pouch is available from:

Cadillac Products
5800 Crooks Road
Troy, Michigan 48098-2830
(248) 813-8200

The filled and sealed activation fluid units are available from:

Heritage Packaging
625 Fishers Run
Victor, NY 14564
(585) 742-3310

4. Heating tray. The heating tray is available from:

Transform Plastics
45 Prince St.
Danvers, MA 01923
(978) 777-1440

Ecomass Technologies
4101 Parkstone Heights Drive
Austin, TX 78746-7482
(512) 306-0020

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5. Pull tab. The pull tab is identified as Part # R16-4-WCSHCE-4. The pull tab is available from:

Southern Imperial
1400 Eddy Avenue
P.O. Box 2308
Rockford, IL 61103
(800) 747-4665 x203

6. Thermocoupled polymeric trays. Filled thermocoupled polymeric trays or instructions on how to construct them are available from:

US Army Research, Development and Engineering Command
Natick Soldier Research, Development and Engineering Center
RDNS-CEG
15 Kansas Street
Natick, MA 01760-5056
(508) 233-4939

7. Heating Tray with Heater. The heater sealed into a heating tray is available from:

Ecomass Technologies
4101 Parkstone Heights Drive
Austin, TX 78746-7482
(512) 306-0020

8. Assembly and instruction sheets. The following assembly and operating instruction sheets are attached:

FIGURE 6. Type II Operating Instructions, for Heater with Hydrogen Generating Warning
FIGURE 7. Type I Operating Instructions, for Heater with No Hydrogen Generation Warning
FIGURE 8. Type II with Eggs Operating Instructions, for Heater with Hydrogen Generating Warning
FIGURE 9. Type II with Eggs Operating Instructions, for Heater with No Hydrogen Generation Warning
FIGURE 10. Assembly Instructions for Type II Heater Module
FIGURE 11. Assembly Instructions for Type I Heater Module with Eggs

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Assembly and instruction sheets in color are available electronically from:

US Army Research, Development and Engineering Command
Natick Soldier Research, Development and Engineering Center
RDNS-CEG
15 Kansas Street
Natick, MA 01760-5056
(508) 233-6252

B. References.

MIL-PRF-131	-	Barrier Materials, Watervaporproof, Greaseproof, Flexible, Heat-Sealable

(Copies of these documents are available from https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Ave, Building 4D, Philadelphia, PA 19111-5094.)

C. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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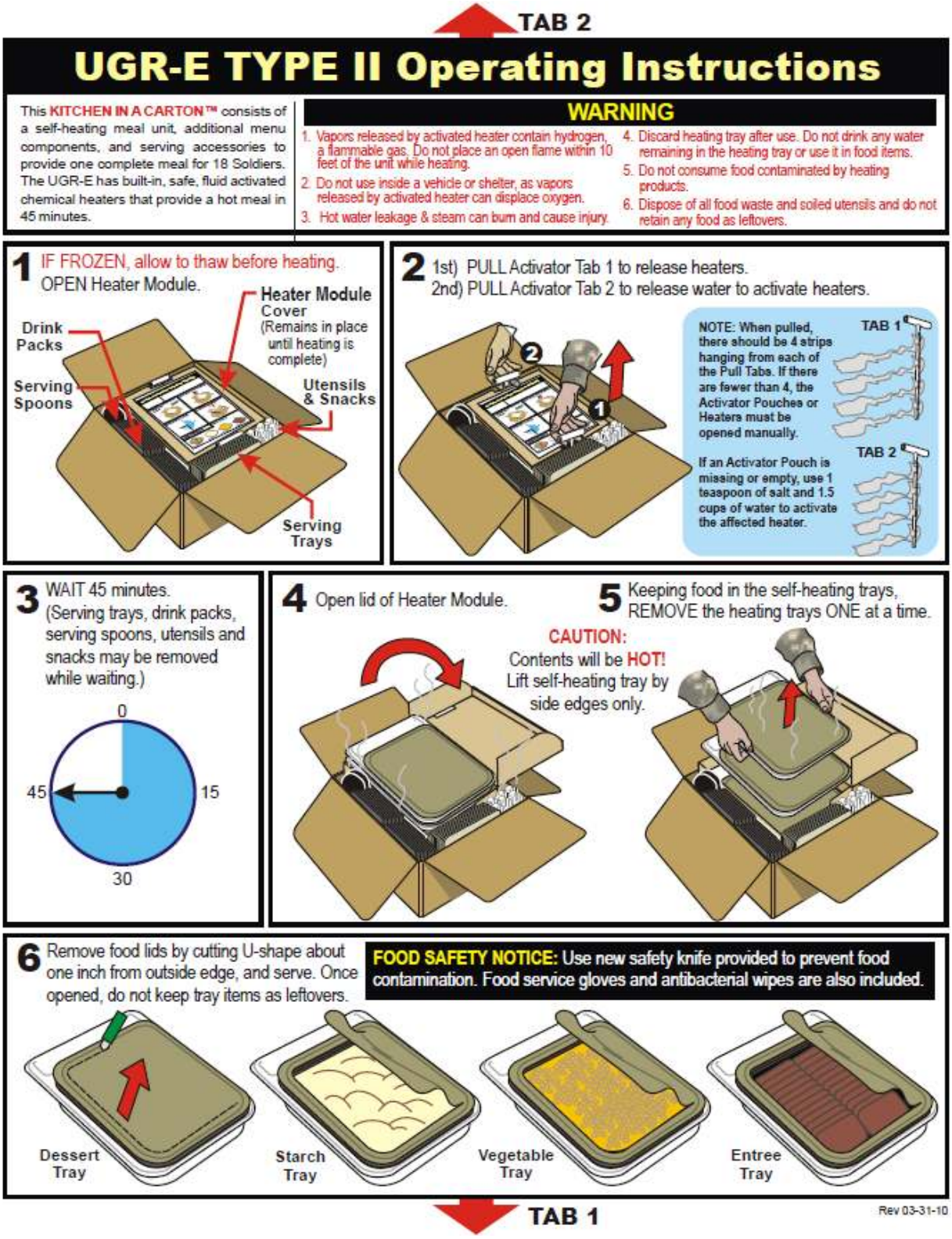


FIGURE 6. Type II Operating Instructions, for Heater with Hydrogen Generating Warning

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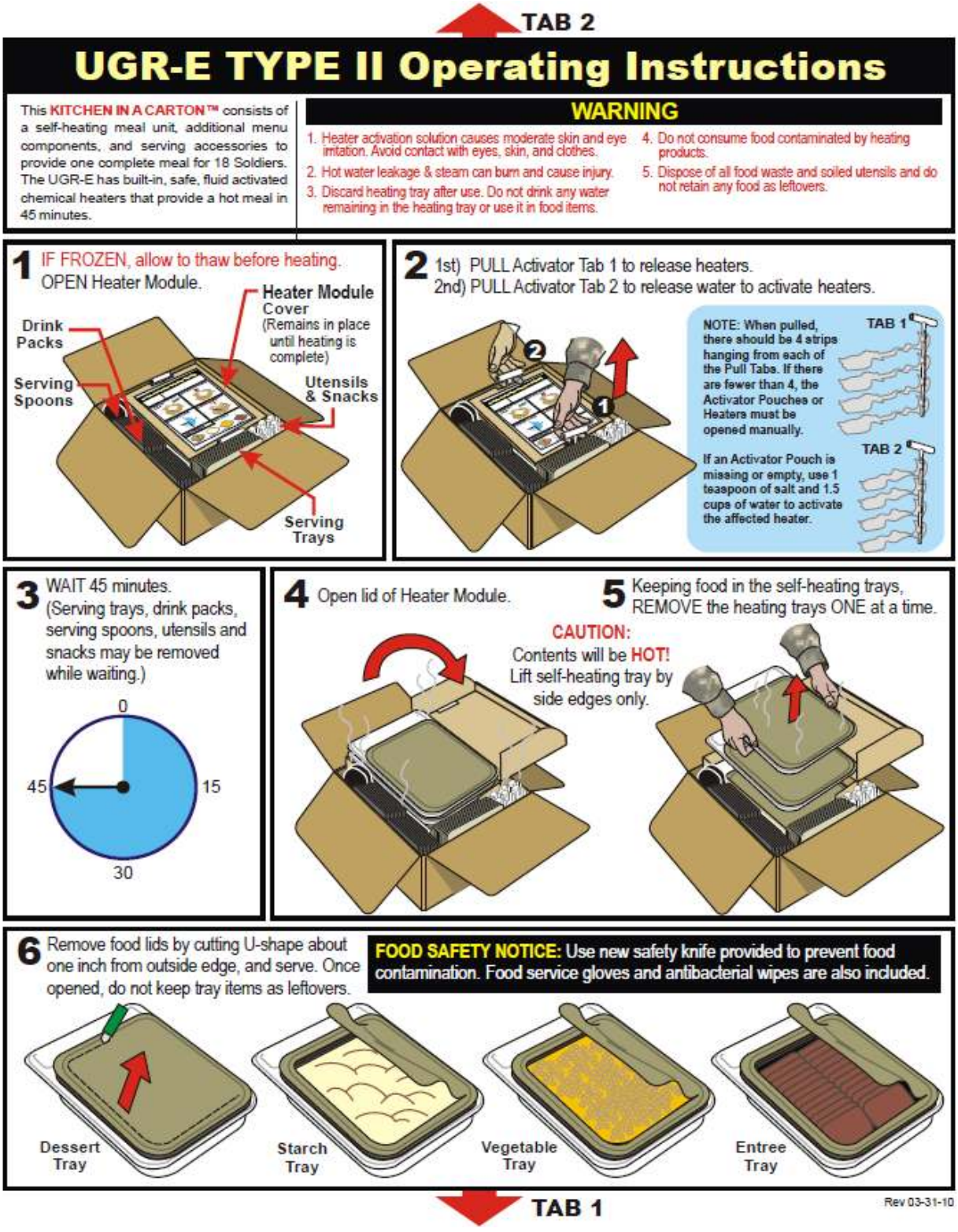


FIGURE 7. Type II Operating Instructions, for Heater with No Hydrogen Generation Warning

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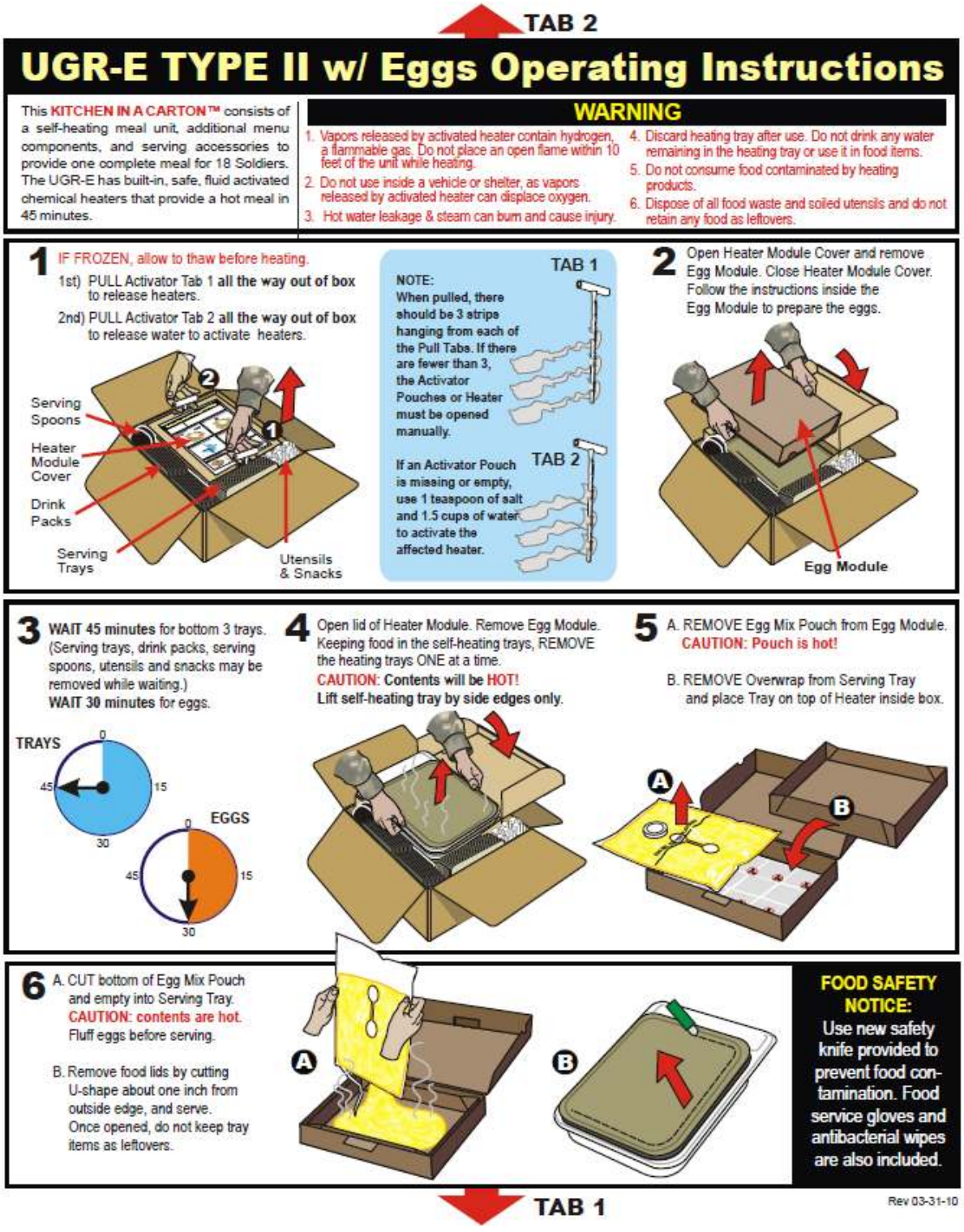


FIGURE 8. Type II with Eggs Operating Instructions, for Heater with Hydrogen Generating Warning

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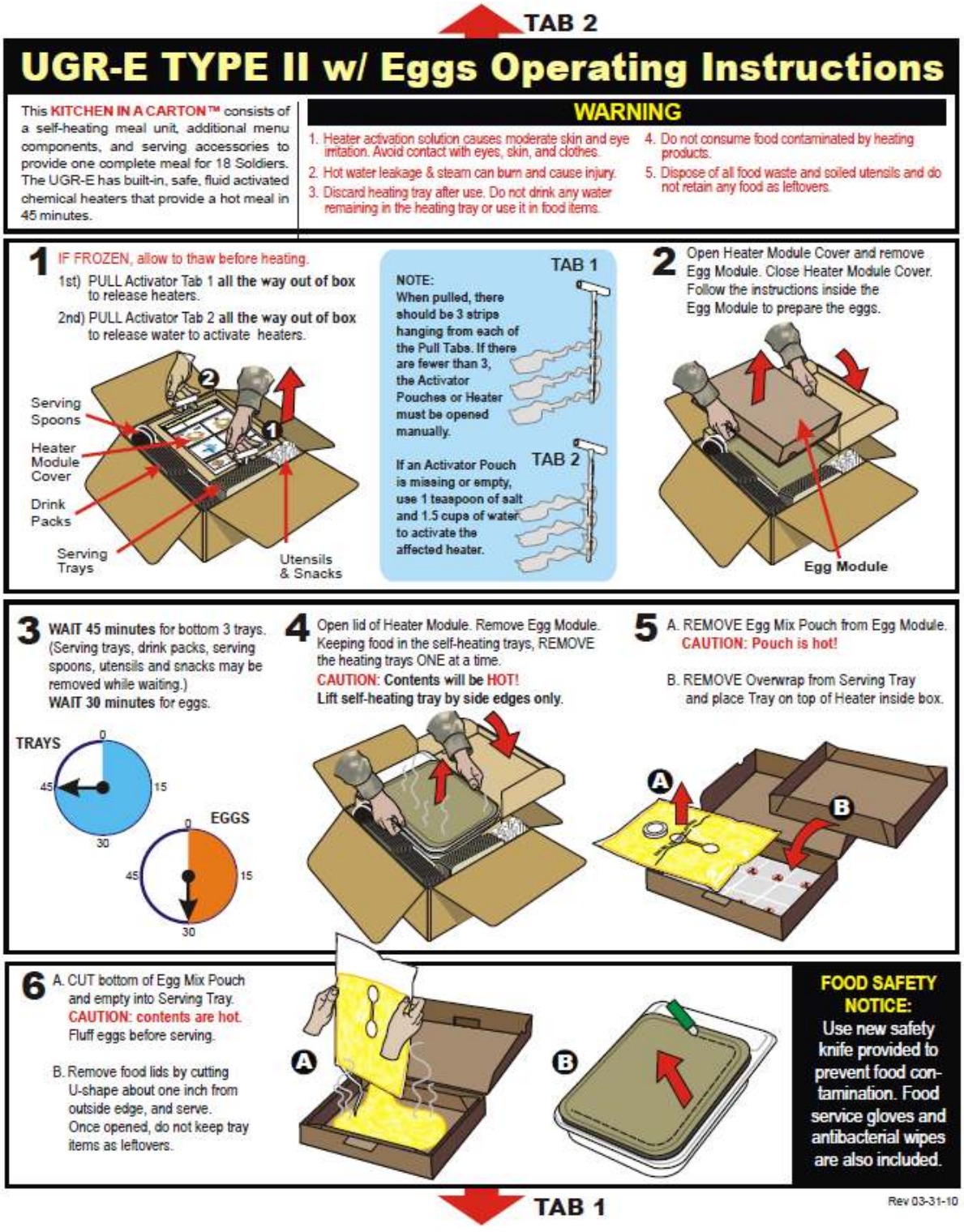


FIGURE 9. Type II with Eggs Operating Instructions, for Heater with No Hydrogen Generation Warning

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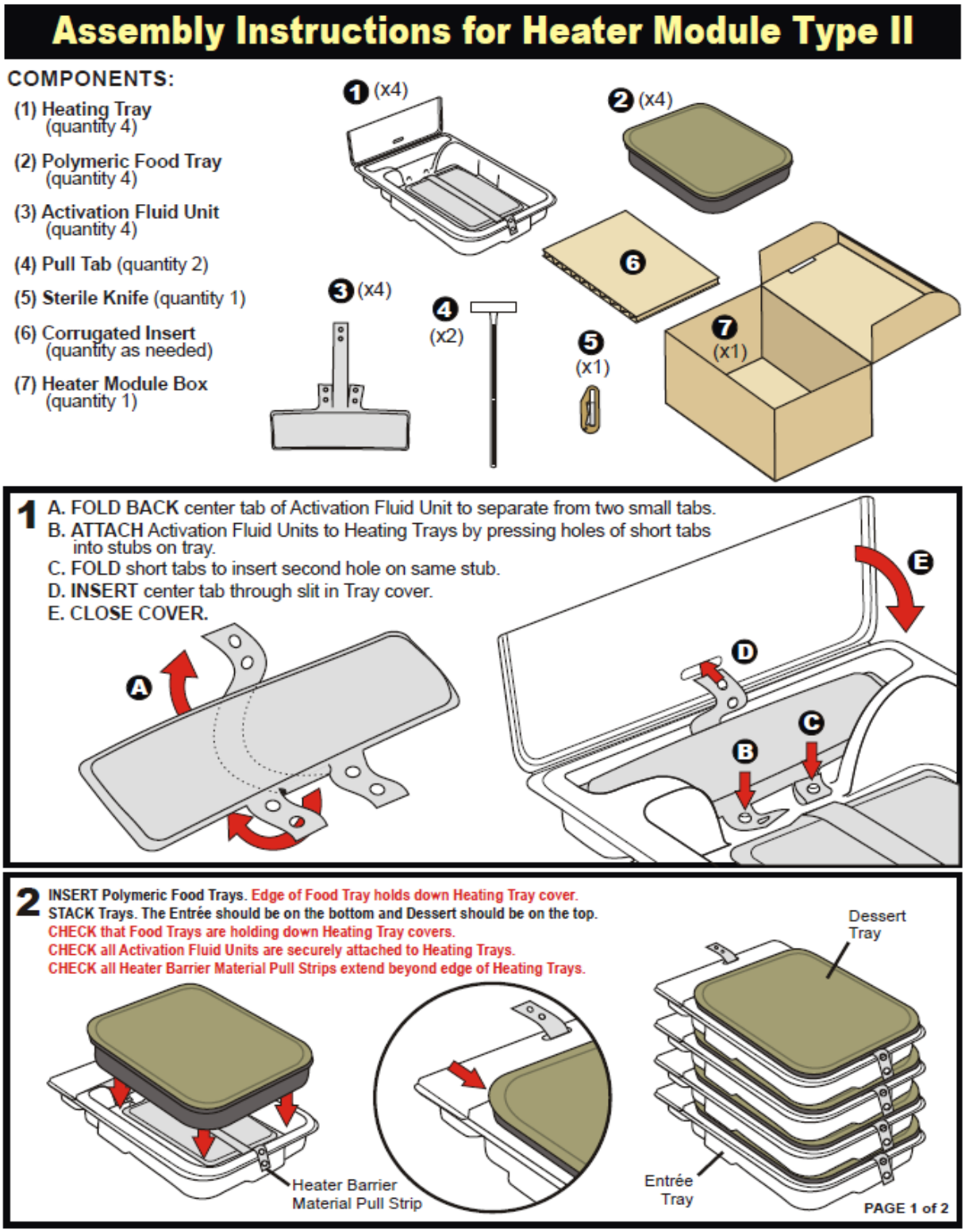


FIGURE 10. Assembly Instructions for Type II Heater Module

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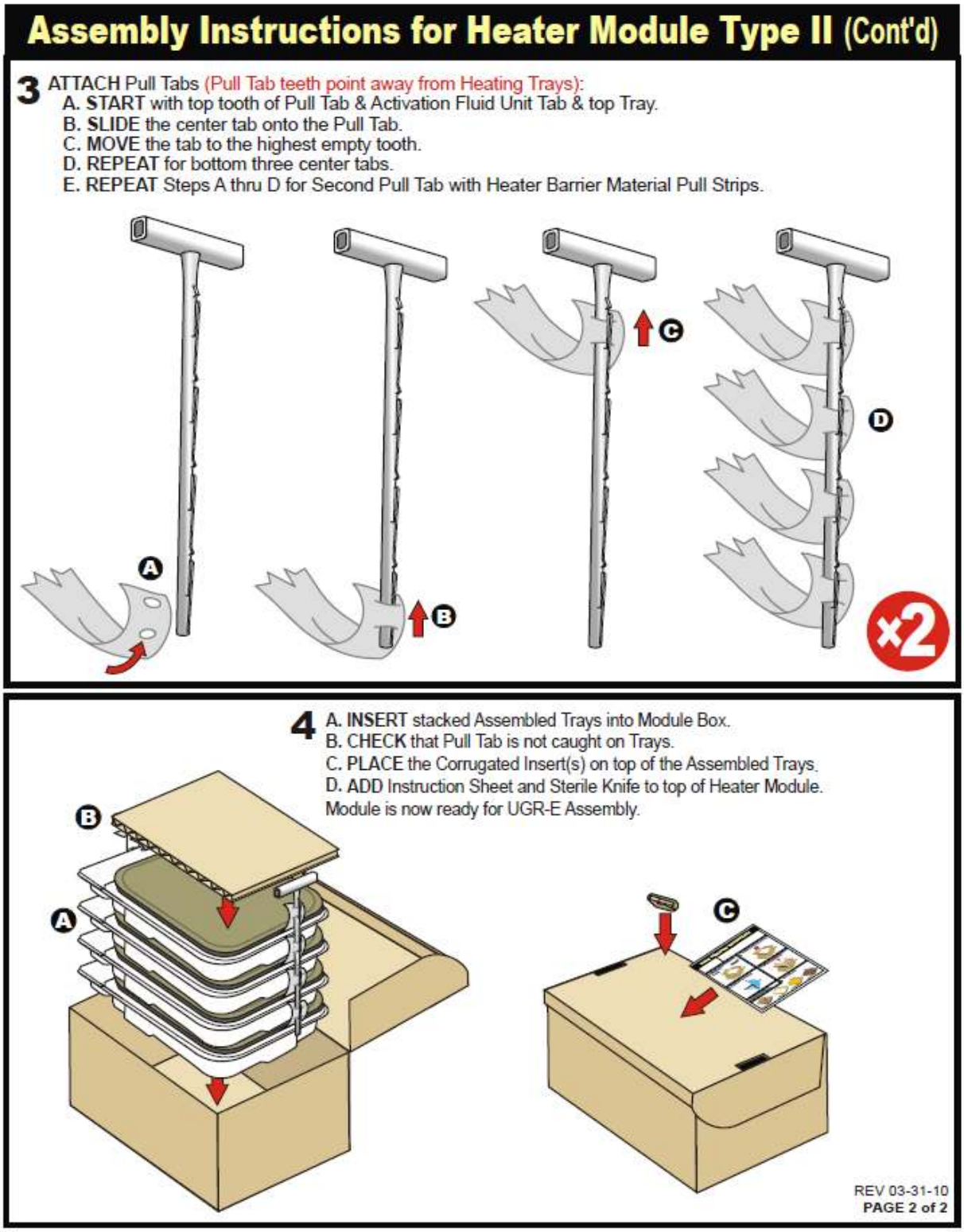


FIGURE 10. Assembly Instructions for Type II Heater Module – Continued

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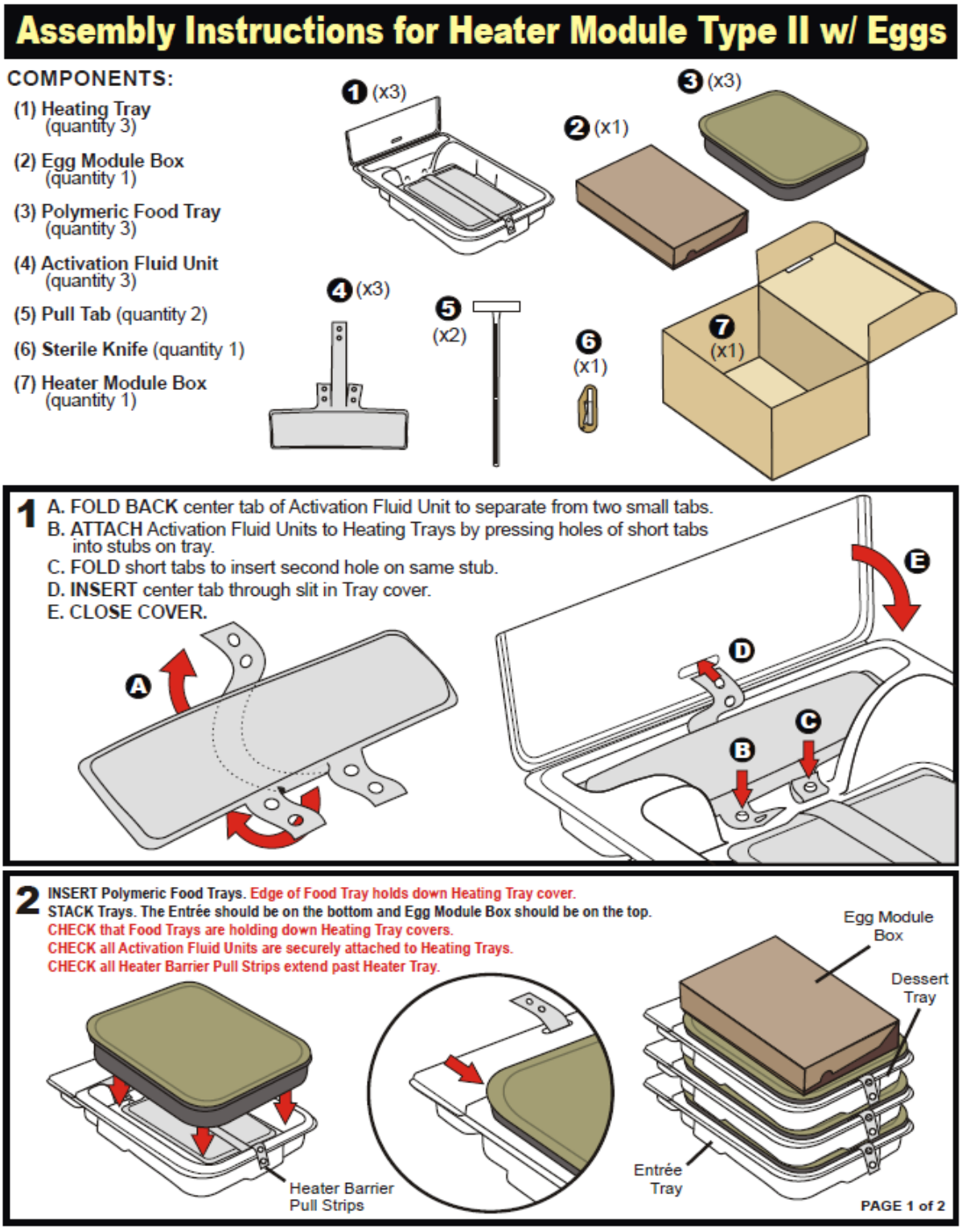


FIGURE 11. Assembly Instructions for Type II Heater Module with Eggs

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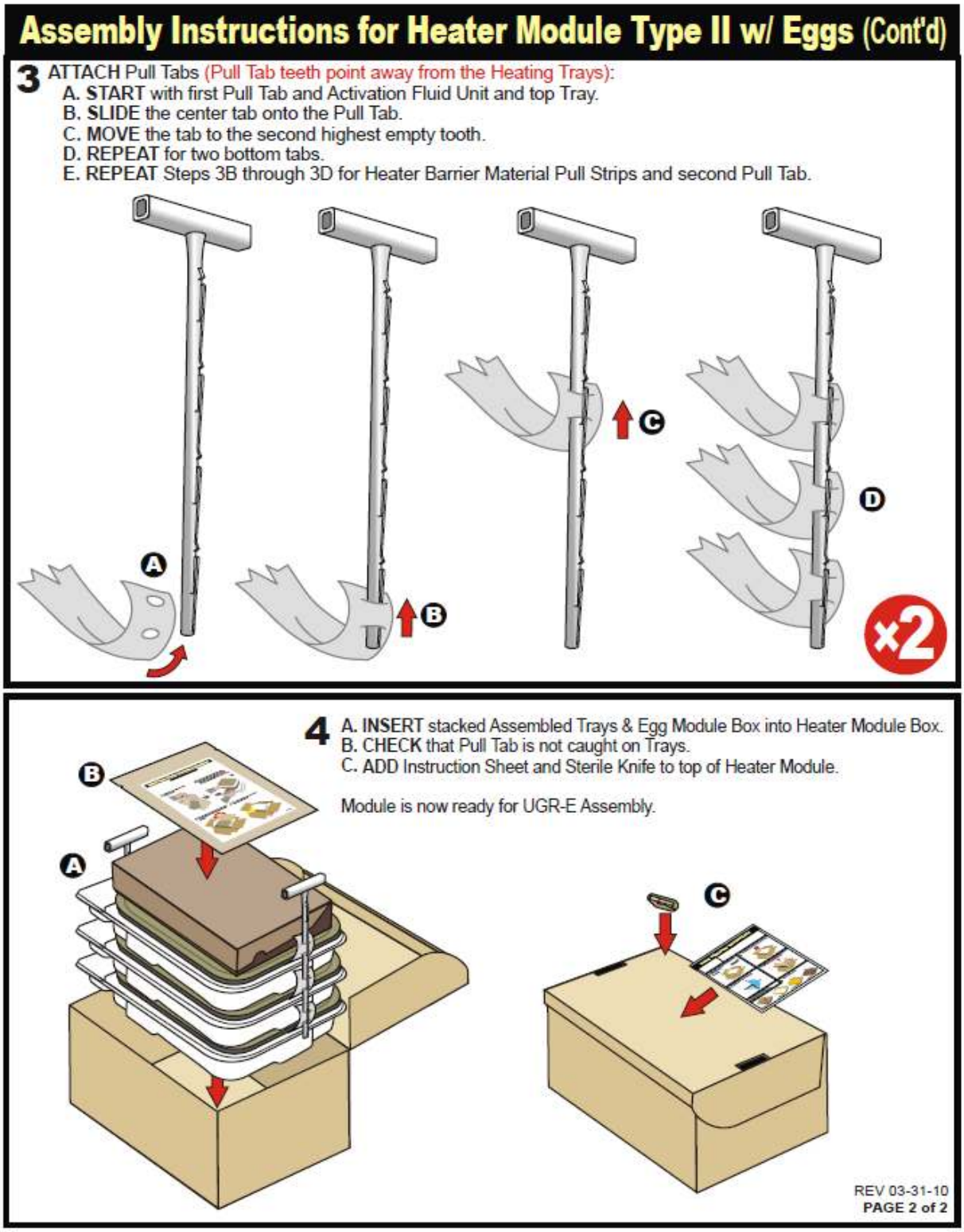


FIGURE 11. Assembly Instructions for Type II Heater Module with Eggs – Continued

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Custodians:
Army – GL
Navy – SA
Air Force – 35

Preparing activity:
Army – GL
(Project 8970-2010-003)

Review Activities:
Army – MD, QM
Navy – MC
DLA – SS

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